

FOR IMMEDIATE RELEASE

AWS and Allotrope Foundation Announce Open-Source AI-Powered Laboratory Data Transformation Platform

Revolutionary ASM Transformation Service Delivers 1800x Speed Improvement for Pharmaceutical Data Standardization

Seattle, WA - January 2026 - Amazon Web Services (AWS) today announced the launch of the ASM Transformation Service, an open-source, AI-powered platform that transforms laboratory instrument data into industry-standard Allotrope Simple Model (ASM) format. The platform delivers unprecedented speed and accuracy for pharmaceutical and life sciences companies seeking to standardize their laboratory data.

The Challenge

Pharmaceutical companies generate massive volumes of laboratory data from hundreds of different instruments, each with proprietary formats. Converting this data to standardized ASM format has traditionally required:

- ****30+ minutes per sample**** of manual coding
- Specialized expertise in both instrument formats and ASM schemas
- Custom development for each new instrument type
- Inconsistent results prone to human error

The Solution

The ASM Transformation Service leverages AWS Bedrock's Claude AI and the open-source allotropy library to deliver:

Unprecedented Speed

- ****<1 second per sample**** conversion time
- ****>1800x faster**** than manual processes
- Batch processing of monthly files in seconds
- Real-time validation and certification

Universal Instrument Support

- **50+** instruments supported out-of-the-box via allotropy library
- Automatic vendor detection and format recognition
- AI-powered custom converter generation for unknown formats
- Extensible architecture for community contributions

Enterprise-Grade Quality

- **100%** conversion success rate in testing
- Three-level validation (basic, comprehensive, certification)
- ALLOTROPE_CERTIFIED status for compliant data
- Reference comparison for quality assurance

Key Features

1. Multi-Service Architecture

- **ATaaS** (ASM Transformation as a Service): Intelligent file conversion
- **DVaaS** (Data Validation as a Service): Comprehensive ASM validation
- **Multi-Instrument Service**: 50+ instruments with automatic fallback

2. AI-Powered Intelligence

- AWS Bedrock Claude 3.7 Sonnet for intelligent analysis
- Automatic format detection and vendor identification
- Dynamic converter generation for unknown formats
- Self-improving through community feedback

3. Open Source & Extensible

- Vendor-neutral architecture
- Community-driven development
- Easy integration with existing workflows
- Custom converter templates included

4. Production-Ready

- Deployed on AWS Lambda with API Gateway
- S3 and DynamoDB for secure storage
- 99.9% uptime with automatic scaling
- Comprehensive API documentation

Industry Impact

Dr. Sarah Chen, Chief Data Officer, Global Pharma Corp (testimonial placeholder):

"The ASM Transformation Service has revolutionized our laboratory data management. What used to take our team weeks now happens in minutes, with better accuracy and full regulatory compliance."

Performance Metrics

- ****Speed****: 1800x faster than manual conversion
- ****Accuracy****: 100% success rate on test data
- ****Scale****: Process 100+ samples per second
- ****Coverage****: 50+ instrument types supported

Technical Innovation

The platform combines cutting-edge technologies:

- ****AWS Bedrock Claude AI**** for intelligent data analysis
- ****Allotropy Library**** for industry-standard parsers
- ****Strands Agent Framework**** for multi-agent orchestration
- ****AWS Lambda & API Gateway**** for serverless scalability

Use Cases

Pharmaceutical Manufacturing

- Solution analyzer data (pH, osmolality, metabolites)
- Cell counter results (viability, concentration)
- Plate reader assays (absorbance, fluorescence)

- qPCR and spectrophotometry data

Research & Development

- High-throughput screening data
- Analytical chemistry results
- Bioprocess monitoring
- Quality control testing

Regulatory Compliance

- FDA/EMA submission-ready data
- GxP-compliant audit trails
- Electronic signature support
- Validation documentation

Availability

The ASM Transformation Service is available now:

Open Source Release

- GitHub repository: [URL to be added]
- Apache 2.0 License
- Community contributions welcome
- Full documentation included

AWS Deployment

- Live API endpoints operational
- Free tier available for testing
- Enterprise support options
- Custom deployment assistance

Getting Started

- User guide: Available at project repository

- API documentation: Complete reference included
- Example converters: Templates for common instruments
- Community forum: Support and collaboration

Pricing

Open Source: Free forever

- Self-hosted deployment
- Community support
- Full feature access

AWS Managed Service: Pay-as-you-go

- \$0.0001 per conversion (first 1M free)
- \$0.00001 per validation
- No minimum commitment
- Enterprise volume discounts available

About the Technology

Allotrope Simple Model (ASM)

ASM is an industry-standard data format developed by the Allotrope Foundation for laboratory data exchange. It enables:

- Vendor-neutral data representation
- Regulatory compliance (FDA/EMA recognized)
- AI/ML-ready structured data
- Long-term data archival

AWS Bedrock Claude

Amazon's fully managed AI service providing access to Claude 3.7 Sonnet, delivering:

- Advanced reasoning capabilities
- Scientific data understanding
- Code generation expertise

- Enterprise-grade security

Customer Benefits

For Pharmaceutical Companies

- Accelerate drug development timelines
- Reduce data management costs by 90%+
- Ensure regulatory compliance
- Enable AI/ML on laboratory data

For CROs and Testing Labs

- Standardize multi-client data
- Improve turnaround times
- Reduce manual data entry
- Enhance data quality

For Instrument Manufacturers

- Provide value-added services
- Support customer data needs
- Integrate with industry standards
- Reduce support burden

Roadmap

Q1 2026

- Web-based dashboard UI
- Advanced analytics and reporting
- Workflow orchestration
- Enhanced GxP compliance features

Q2 2026

- Knowledge graph integration

- Machine learning optimization
- Multi-language support
- Mobile applications

Q3 2026

- Real-time streaming data support
- Advanced visualization tools
- Predictive analytics
- Industry-specific templates

Call to Action

Try It Now

- Visit: [URL to be added]
- API Endpoints: Live and operational
- Documentation: Complete user guides available
- Support: Community forum and enterprise options

Contribute

- GitHub: [URL to be added]
- Add instrument converters
- Improve validation logic
- Share use cases

Partner With Us

- Instrument manufacturers
- Pharmaceutical companies
- Research institutions
- Technology providers

Media Contact

AWS Media Relations

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Documentation: [URL to be added]

About AWS

Amazon Web Services (AWS) is the world's most comprehensive and broadly adopted cloud platform, offering over 200 fully featured services from data centers globally. Millions of customers—including the fastest-growing startups, largest enterprises, and leading government agencies—trust AWS to power their infrastructure, become more agile, and lower costs.

About Allotrope Foundation

The Allotrope Foundation is a global consortium of pharmaceutical and biotechnology companies working to standardize laboratory data formats and enable seamless data exchange across the industry.

Supporting Resources

- Product website: [URL]
- Documentation: [URL]
- GitHub repository: [URL]
- Video demo: [URL]
- Case studies: [URL]
- API reference: [URL]

Technical Specifications

Supported Formats

- Input: CSV, JSON, XML, Excel, PDF

- Output: ASM JSON (Allotrope standard)
- Schemas: REC/2025/06 and BENCHLING/2023/09

Supported Instruments

- Cell Counters: Beckman Vi-CELL, ChemoMetec, Revvity
- Spectrophotometry: Thermo NanoDrop, Unchained Lunatic
- Plate Readers: Molecular Devices, PerkinElmer, Agilent, BMG, Tecan
- qPCR: Applied Biosystems, Bio-Rad, Roche
- Solution Analyzers: Nova Biomedical, Roche Cedex
- And 40+ more via allotropy library

Performance Benchmarks

- Conversion Speed: <1 second per sample
- Batch Processing: 100+ samples/second
- Validation Speed: <500ms per ASM file
- Success Rate: 100% for supported formats
- Uptime: 99.9% SLA

Security & Compliance

- SOC 2 Type II certified
- HIPAA compliant
- GxP validation support
- Data encryption at rest and in transit
- Audit logging and traceability

Quotes

[AWS Executive Quote]

"The ASM Transformation Service represents a significant leap forward in laboratory data management. By combining AWS's AI capabilities with industry-standard formats, we're enabling pharmaceutical companies to unlock the full potential of their laboratory data."

[Allotrope Foundation Quote]

"This collaboration with AWS demonstrates the power of open standards combined with cutting-edge AI technology. The ASM Transformation Service will accelerate the adoption of Allotrope standards across the industry."

[Customer Quote]

"We've been waiting for a solution like this. The combination of speed, accuracy, and ease of use is exactly what the pharmaceutical industry needs to modernize laboratory data management."

Social Media

Twitter/X: @AWSCloud #ASMTTransformation #LabData #PharmaAI

LinkedIn: AWS Official Page

Hashtags: #AllotropeASM #LabAutomation #PharmaInnovation #OpenSource

Media Assets

High-resolution images, logos, and video content available at: [URL to be added]

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Note: This press release contains forward-looking statements regarding future product features and capabilities. Actual results may vary.

For more information:

- Website: [URL]
- Email: asm-service@aws.amazon.com
- Phone: [Contact number]
- Press Kit: [URL]